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B.Tech. Degree IV Semester Examination April 2018

CE/CS/EC/EE/IT/ME/SE

AS 15-1401 COMPLEX VARIABLES AND PARTIAL DIFFERENTIAL EQUATIONS
(2015 Scheme)

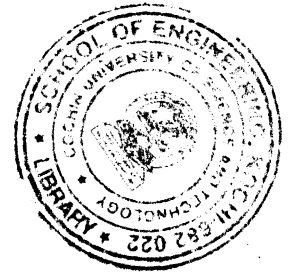
Time : 3 Hours

Maximum Marks : 60

PART A
(Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Prove the polar form of Cauchy Riemann Equations, $\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta}$ and $\frac{\partial v}{\partial r} = -\frac{1}{r} \frac{\partial u}{\partial \theta}$.
- (b) Prove that $v = \tan^{-1}(y/x)$ is harmonic.
- (c) Find the bilinear transform which maps the points $z: 1, i, -1$ into the points $w: 0, 1, \infty$.
- (d) Evaluate $\int_0^{2+3i} (x^2 - iy) dz$ along the path $y = x^2$.
- (e) State and prove Cauchy's Integral formula.
- (f) Evaluate $\int_C \frac{e^{2z} dz}{(z+1)^4}$, $C: |z| = 2$.
- (g) Form partial differential equation from $z = f(x+ct) + \phi(x-ct)$.
- (h) Solve $p(1+q) = qz$.
- (i) Using method of separation of variables solve $\frac{\partial u}{\partial x} = 2 \frac{\partial u}{\partial t} + u$.
- (j) Find the solution of one dimensional heat equation.



PART B

(4 × 10 = 40)

- II. (a) Show that the function $f(z) = \sqrt{xy}$ is not analytic at the origin even though Cauchy-Riemann equations are satisfied at the origin.
- (b) Let $w = u + iv$ be an analytic function. Then prove that the family of curves $u(x, y) = c_1$ and $v(x, y) = c_2$ cut orthogonally.

OR

- III. (a) Show that the function $f(z) = \sinh z$ is an analytic function and find its derivative.
- (b) Find the image of $|z - 2i| = 2$ under the mapping $w = \frac{1}{z}$.

(P.T.O.)

IV. (a) Using Cauchy's Integral Formula, evaluate $\int_C \frac{z}{(z-1)(z-2)^2} dz$, $C: |z-2| = \frac{1}{2}$.

(b) Expand using Laurent's series :

(i) $f(z) = \frac{1}{z(z-1)}$, $0 < |z| < 1$

(ii) $f(z) = \frac{1}{z^2 - 3z - 2}$, $0 < |z-1| < 1$

OR

V. (a) Using Cauchy's Residue theorem, evaluate $\int_C \frac{3z^2 + z - 1}{(z^2 - 1)(z - 3)} dz$, $C: |z| = 2$.

(b) Using Contour integration, evaluate $\int_0^{2\pi} \frac{\cos \theta}{3 + \sin \theta} d\theta$.

VI. Solve

(i) $z^2(p^2 + q^2 + 1) = b^2$

(ii) $p^2 + q^2 = z^2(x + y)$

OR

VII. Solve

(i) $x^2(y-z)p + y^2(z-x)q - z^2(x-y) = 0$

(ii) $(D^3 - 3D^2D^1 + 4D^1^3)z = e^{x+2y}$

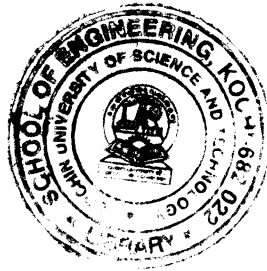
VIII. (a) Derive one dimensional wave equation.

(b) Obtain solution of Laplace equation by the method of separation of variables.

OR

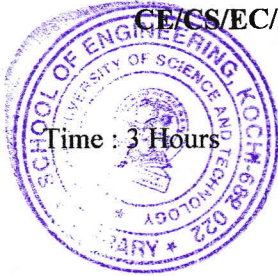
IX. A tightly stretched string with fixed end points $x=0$ and $x=l$ is initially in the position $y=f(x)$. It is vibrating by giving each of its points a velocity

$\frac{\partial y}{\partial t} = g(x)$ at $t=0$. Find $y(x, t)$ in the form of Fourier Series.



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B.Tech. Degree IV Semester Examination April 2019



**CE/CS/EC/EE/IT/ME/SE AS 15-1401 COMPLEX VARIABLES AND PARTIAL
DIFFERENTIAL EQUATIONS
(2015 Scheme)**

Time : 3 Hours

Maximum Marks : 60

PART A
(Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) If $f(z)$ is an analytic function with constant modulus, show that $f(z)$ is constant.
- (b) Find the orthogonal trajectories of the family of curves $r^2 \cos 2\theta = c$.
- (c) Show that the transformation $w = \frac{1}{z}$ transforms all circles and straight lines into circles and straight lines in the w -plane. Which circles in the z -plane become straight lines in the w -plane and which circles are transformed into other circles?
- (d) Evaluate $\int_c \frac{z}{z^2 - 3z + 2} dz$ where c is the circle $|z+1| = 1$.
- (e) Determine the poles and the residue at each pole of $\frac{z^2 + 1}{z^2 - 2z}$.
- (f) Find the Taylor's series expansion of $f(z) = \frac{1}{z^2 - z - 6}$ about $z = 1$.
- (g) Form the partial differential equation by eliminating the arbitrary function from $xyz = f(x + y + z)$.
- (h) Solve $yp + xq + pq = 0$.
- (i) Find the solution of Laplace equation over a rectangular region by the method, of separation of variables.
- (j) Using D'Alembert's method find the deflection of a vibrating string of unit length having fixed ends with initial velocity zero and initial deflection $f(x) = a(x - x^2)$.

PART B

(4 × 10 = 40)

- II. (a) Show that the function $f(z) = \sqrt{|xy|}$ is not regular at the origin, although C-R equations are satisfied.
- (b) Find the analytic function whose real part is $e^{-x}(x \sin y - y \cos y)$.

OR

- III. (a) Determine the linear fractional transformation that sends the points $z = 0, -1, 2i$ into the points $w = 5i, \infty, -\frac{i}{3}$ respectively. Find the image of $|z| < 1$ under this transformation.
- (b) Under the transformation $w = \frac{z-i}{1-iz}$, find the map of the circle $|z| = 1$ in the w plane.

(P.T.O.)

IV. (a) Evaluate $\int_c \frac{e^z}{(z-i)^2(z^2+4)} dz$ where $c: |z-1|=1/2$, using Cauchy's integral formula.

(b) Evaluate $\int_0^{2\pi} \frac{\cos 3\theta}{5-4\cos \theta} d\theta$.

OR

V. (a) Find the Laurent expansion of $\frac{7z^2+9z-18}{z^3-9z}$ in the region (i) $0 < |z| < 3$
(ii) $|z| > 3$.

(b) Evaluate $\int_c \frac{dz}{(z^2+4)^2}$ where $c: |z-i|=2$, using residue theorem.

VI. (a) Solve $x^2(y-z)p + y^2(z-x)q = z^2(x-y)$.

(b) Solve $z^2(p^2+q^2) = x^2+y^2$.

OR

VII. (a) Solve $\frac{\partial^3 z}{\partial x^3} - 2\frac{\partial^3 z}{\partial x^2 \partial y} = 2e^{2x} + 3x^2 y$.

(b) Solve $(D^2 + DD' - 6D'^2)z = \cos(2x+y)$.

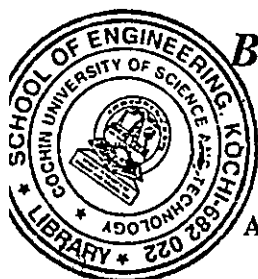
VIII. (a) Derive one dimensional heat equation.

(b) A tightly stretched string of length 1 with fixed ends is initially in equilibrium position. It is set vibrating by giving each point a velocity $V_0 \sin^3(\frac{\pi x}{\ell})$. Find the displacement $y(x,t)$.

OR

IX. (a) A bar 100 cm long, with insulated sides, has its ends kept at 0°C and 100°C until steady state conditions prevail. The two ends are then suddenly insulated and kept so. Find the temperature distribution.

(b) Solve $u_{xx} + u_{yy} = 0$ which satisfies the conditions $u(0,y) = u(l,y) = u(x,0) = 0$
and $u(x,a) = \sin \frac{n\pi x}{l}$.



B.Tech. Degree IV Semester Supplementary Examination April 2021

CE/CS/EC/EE/IT/ME/SE-

AS 15-1401 COMPLEX VARIABLES AND PARTIAL DIFFERENTIAL EQUATIONS
(2015 Scheme)

Time: 3 Hours

Maximum Marks: 60

PART A (Answer ALL questions)

(10 × 2 = 20)

- I.
 - (a) Verify Cauchy Riemann equations for an analytic function.
 - (b) Prove that real and imaginary parts of an analytic function are harmonic.
 - (c) Test the analyticity of the function $f(z) = \bar{z}$.
 - (d) Evaluate $\int_C \frac{\sin(3z)}{z + \frac{\pi}{2}} dz$ if C is the circle $|z| = 5$.
 - (e) Show that $\frac{1}{z^2} = \frac{1}{4} \sum_{n=0}^{\infty} (-1)^n (n+1) \left(\frac{z-2}{2}\right)^n$ when $|z-2| < 2$.
 - (f) Form the partial differential equation by eliminating the arbitrary function g from the relation $g(x+y+z, x^2+y^2+z^2) = 0$.
 - (g) Solve $z = p^2 + q^2$.
 - (h) Solve $q^2 - p = y - x$.
 - (i) Using the method of separation of variables solve $\frac{\partial u}{\partial x} = 2 \frac{\partial u}{\partial t} + u$ where $u(x, 0) = 6e^{-3x}$.
 - (j) Derive one dimensional wave equation.

PART B

(4 × 10 = 40)

- II.
 - (a) Find an analytic function whose imaginary part is $3x^2y - y^3$ also find its harmonic conjugate.
 - (b) Find the image of the circle $|z-1|=1$ in the complex plane under the mapping $w = \frac{1}{z}$.

OR

- III.
 - (a) Show that $w = \sin z$ transform the semi infinite strip $0 \leq x \leq \frac{\pi}{2}, y \geq 0$ ONTO the first quadrant of the w -plane.
 - (b) Find the bilinear transformation which maps the points $z_1 = 2, z_2 = i$ and $z_3 = -2$ into the points $w_1 = 1, w_2 = i$ and $w_3 = -1$.

(P.T.O.)

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IV.

(a) Using Cauchy's integral formula, find the value of $\int_C \frac{z+4}{z^2+2z+5} dz$ where C is the circle $|z+1-i|=2$.

(b) Find the residue of $f(z) = \frac{\sin(z)}{z \cos(z)}$ at each of its poles inside the circle $|z|=2$.

OR

V.

(a) Using residue theorem, evaluate $\int_C \frac{dz}{(z^2+4)^2}$ where C is the circle $|z-i|=2$.

(b) Using contour integration, evaluate $\int_0^{2\pi} \frac{d\theta}{2+\cos(\theta)}$.

VI.

Solve

(i) $x^2 p^2 + y^2 q^2 = z^2$

(ii) $(x-y)p + (y-x-z)q = z$

OR

VII.

Solve

(i) $(D^2 - 2DD')(z) = e^{2x} + x^3 y$

(ii) $(D^2 + DD' - 6D^2)(z) = y \cos(x)$

VIII.

(a) Derive one dimensional heat equation.

(b) Obtain the solution of the Laplace equation by the method of separation of the variables.

OR

IX.

(a) A string is stretched and fastened to two points $x=0$ and $x=l$ apart. Motion is started by displacing the string in to the form $y=k(lx-x^2)$ from which it is released at time $t=0$. Find the displacement of any point on the string at a distance x from one end at time t in the form of Fourier series.

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B.Tech. Degree IV Semester Examination April 2017

CE/CS/EC/EE/IT/ME/SE

AS 15-1401 COMPLEX VARIABLES AND PARTIAL DIFFERENTIAL EQUATIONS
(2015 Scheme)

Time: 3 Hours

Maximum Marks: 60

PART A

(Answer ALL questions)

(10 × 2 = 20)

- I. (a) Test whether the function $f(z) = xy + iy$ analytic or not.
- (b) Find an analytic function whose imaginary part is $3x^2y - y^3$.
- (c) Find the image of the circle $|z - 1| = 1$ in the complex plane under the mapping $w = \frac{1}{z}$.
- (d) Show that cross ratio remains invariant under a bilinear transformation.
- (e) Show that (i) $\frac{1}{z^2} = \sum_{n=0}^{\infty} (n+1)(z+1)^n$ when $|z+1| < 1$ and
- (ii) $\frac{1}{z^2} = \frac{1}{4} \sum_{n=0}^{\infty} (-1)^n (n+1) \left(\frac{z-2}{2}\right)^n$ when $|z-2| < 2$
- (f) Form the partial differential equation of all spheres having centres on the z-axis.
- (g) Solve $3p^2 - 2q^2 = 4pq$.
- (h) Solve $z = p^2 + q^2$.
- (i) Derive one dimensional heat equation.
- (j) Solve the equation $\frac{\partial u}{\partial x} = 4 \frac{\partial u}{\partial y}$ where $u(0, y) = 8e^{-3y}$ by the method of separation of variables.

PART B

(4 × 10 = 40)

- II. (a) Evaluate using Cauchy's integral formula $\int_c \frac{\sin 3z}{z + \frac{\pi}{2}} dz$ if c is the circle $|z| = 5$.
- (b) Find Laurent's series expansion of $\frac{1}{(z+1)(z+3)}$ in powers of $(z+1)$ for the range $0 < |z+1| < 2$.

OR

(P.T.O.)

III. (a) Evaluate using residue theorem $\int_c \frac{z \sec z}{(1-z^2)} dz$ where c is the ellipse $4x^2 + 9y^2 = 9$.

(b) Evaluate $\int_0^{2\pi} \frac{d\theta}{2 + \cos \theta}$.

IV. (a) Show that the polar form of Cauchy-Riemann equations are $\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta}$, $\frac{\partial v}{\partial r} = -\frac{1}{r} \frac{\partial u}{\partial \theta}$. Deduce that $\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} = 0$.

(b) Discuss the mapping properties of $\omega = z^2$.

OR

V. (a) Find the bilinear transformation which maps the points $z = 1, i, -1$ ONTO the points $\omega = i, 0, -i$.

(b) Show that the function $f(z) = \sqrt{|xy|}$ is not analytic at the origin even though Cauchy-Riemann equations are satisfied at the origin.

VI. (a) Form the partial differential equation by eliminating the arbitrary function from $\phi(x^2 + y^2 + z^2, xyz) = 0$.

(b) Find the general solution of $x(z^2 - y^2)p + y(x^2 - z^2)q = z(y^2 - x^2)$.

OR

VII. (a) Solve $(D^2 - 5DD' + 6D'^2)z = e^{x+y}$.

(b) Solve $q^2 - p = y - x$.

VIII. (a) Solve one dimensional wave equation.

(b) Obtain solution of Laplace's equation over a rectangular region by the method of separation of variables.

OR

IX. (a) Obtain D'Alembert's solution of wave equation.

(b) An insulated rod of length l has its ends A and B maintained at 0°C and 100°C respectively until steady state condition prevail. If B is suddenly reduced to 0°C and maintained at 0°C , find the temperature at a distance x from A at time t .

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**B.Tech. Degree IV Semester Special Supplementary Examination
February 2020**

CE/CS/EC/EE/IT/ME/SE

**AS 15-1401 COMPLEX VARIABLES AND PARTIAL DIFFERENTIAL EQUATIONS
(2015 Scheme)**

Time : 3 Hours

Maximum Marks : 60

PART A
(Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Define analytic function. Give an example of a function which is analytic and an example which is not analytic.
- (b) Show that an analytic function with constant modulus is constant.
- (c) Find the Taylor series expansion of $f(z) = \frac{z-1}{z+1}$ about the point $z = 1$.
- (d) Find the residue of $f(z) = \frac{z^2}{(z-1)^2(z+2)}$ at each of the poles.
- (e) Evaluate $\int_C \frac{dz}{z^2 e^z}$ where C is $|z| = 1$.
- (f) Form the partial differential equation by eliminating ' f ' from the relation $z = y^2 + 2f\left(\frac{1}{x} + \log y\right)$.
- (g) Solve $z = p^2 + q^2$.
- (h) Solve $q^2 - p = y - x$.
- (i) Using the method of separation of variables solve $\frac{\partial u}{\partial x} + 4 = \frac{\partial u}{\partial t}$ with $u = 4e^{-3x}$ when $t = 0$.
- (j) Derive one dimensional wave equation.



PART B

(4 × 10 = 40)

- II. (a) Find the image of the circle $|z-1|=1$ in the complex plane under the mapping $w = \frac{1}{z}$.
- (b) Find the bilinear transformation which maps the points $z = 2, i, -2$ into the points $w = 1, i, -1$.

OR

- III. (a) Find the analytic function $f(z) = u + iv$ if $u + v = \frac{x}{x^2 + y^2}$ and $f(1) = 1$.
- (b) Show that $u = e^x \cos y + xy$ is harmonic and find its harmonic conjugate.

(P.T.O.)

IV. (a) Expand $\frac{1}{(z-1)(z-2)}$ in Laurent's series valid for $|z| < 1$ and $1 < |z| < 2$.

(b) Evaluate $\int_0^{2\pi} \frac{d\theta}{2 + \cos \theta}$.

OR

V. (a) Evaluate $\int_C \frac{dz}{(z^2 + 4)^2}$ where C is the circle $|z - i| = 2$ using residue theorem.

(b) State and prove Cauchy's integral formula.

VI. (a) Find the partial differential equation of all spheres of fixed radius having their centers in the xy plane.

(b) Solve $x^2 p^2 + y^2 q^2 = z^2$.

OR

VII. Solve the following:

(a) $r + s - 6t = \cos(2x + y)$.

(b) $x(y - z)p + y(z - x)q = z(x - y)$.

VIII. (a) Obtain D'Alembert's solution of wave equation.

(b) Derive one dimensional heat equation.

OR

IX. (a) Solve the equation $\frac{\partial u}{\partial t} = \alpha^2 \frac{\partial^2 u}{\partial x^2}$ subject to the boundary condition

$$u(0, t) = 0, u(\ell, t) = 0, u(x, 0) = x.$$

(b) Obtain the solution of Laplace's equation by the method of separation of variables.
